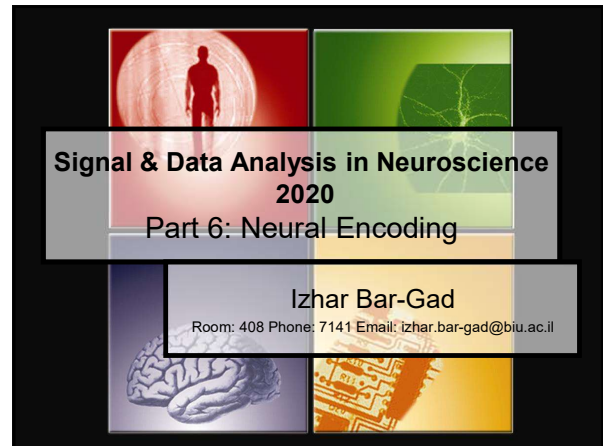
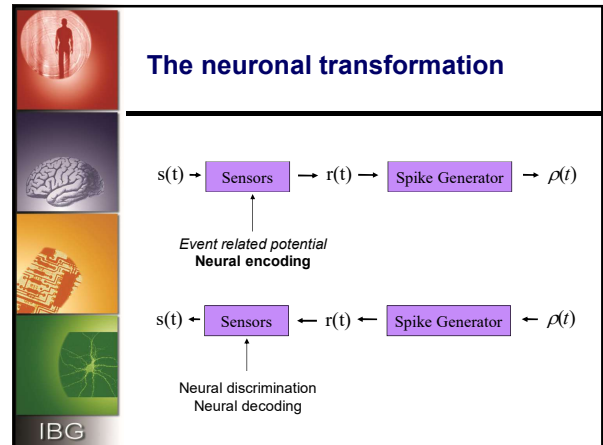



Signal & Data Analysis in Neuroscience
2020
Part 6: Neural Encoding

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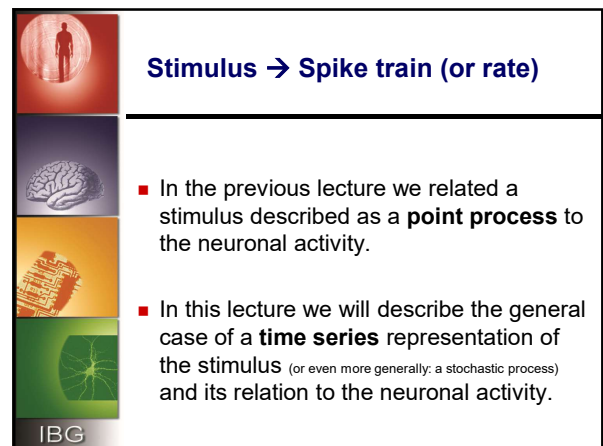
The neuronal transformation

$s(t) \rightarrow \text{Sensors} \rightarrow r(t) \rightarrow \text{Spike Generator} \rightarrow \rho(t)$
 Event related potential
 Neural encoding

$s(t) \leftarrow \text{Sensors} \leftarrow r(t) \leftarrow \text{Spike Generator} \leftarrow \rho(t)$
 Neural discrimination
 Neural decoding

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Stimulus → Spike train (or rate)

- In the previous lecture we related a stimulus described as a **point process** to the neuronal activity.
- In this lecture we will describe the general case of a **time series** representation of the stimulus (or even more generally: a stochastic process) and its relation to the neuronal activity.

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Outline

- Reverse correlations
- Linear filters
- Static non-linearity
- GLMs
- Appendix: V1 simple neurons

Based heavily on:
Theoretical neuroscience, *Dayan P & Abbott LF*, Chapter 2

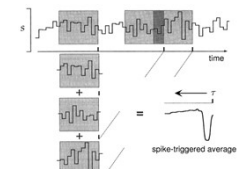
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Spike triggered average

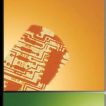
- The interesting question:
What does the spike encode?
- A surrogate question:
What is the average stimulus preceding a spike?



Stimulus and response are defined periodically: $r(t+T)=r(t)$, $s(t+T)=s(t)$.

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Reverse correlation

- Average stimulus preceding the spike → Reverse correlation of the stimulus and the spike train

$$C(\tau) = \frac{1}{n} \sum_{i=1}^n s(t_i - \tau)$$

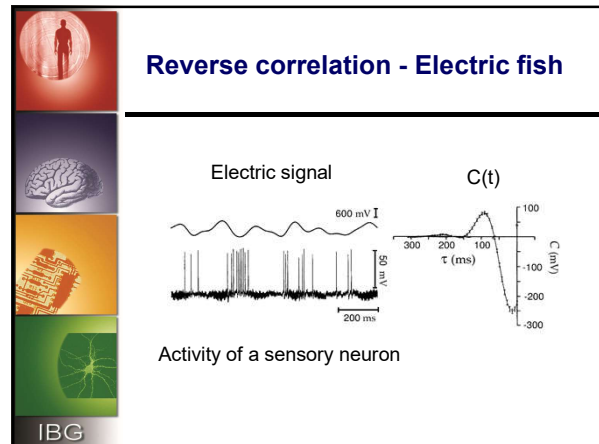
$$= \frac{1}{n} \int dt \rho(t) s(t - \tau)$$

$$\langle C(\tau) \rangle = \frac{1}{n} \int dt r(t) s(t - \tau)$$

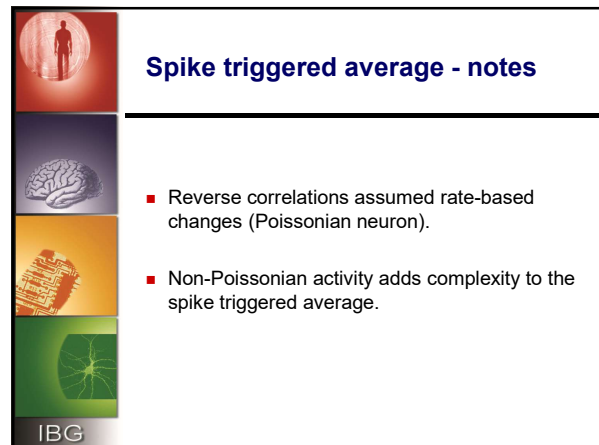
$$= \frac{1}{r} Q_{rs}(-\tau)$$

with $r = n/T$ the average spike rate and $Q_{rs} = \frac{1}{T} \int_0^T dt r(t) s(t + \tau)$ the correlation between the signals r and s .

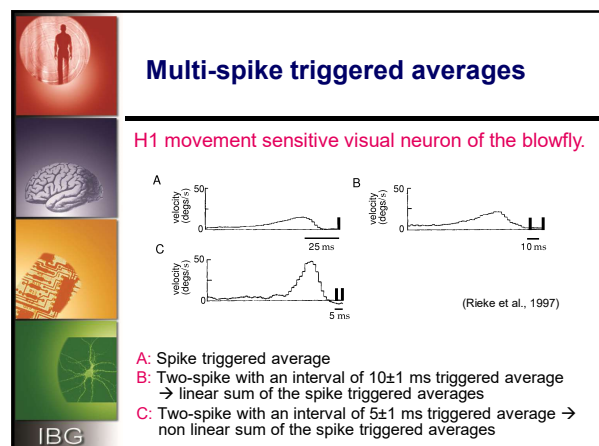
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Outline

- Reverse correlations
- **Linear filters**
- Static non-linearity
- GLMs
- Appendix: V1 simple neurons

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Stimulus → Rate

- We will try to describe the rate of the neuron as a function of the stimulus at all previous time.
- We will define this function as a filter on the stimulus. Filters will be discussed in great detail in the spectral part of course...
- The general description (Volterra series)

$$r_{est}(t) = r_0 + \int_0^{\infty} d\tau D(\tau) s(t - \tau) + \int_0^{\infty} d\tau_1 \int_0^{\infty} d\tau_2 D_2(\tau_1, \tau_2) s(t - \tau_1) s(t - \tau_2) + \dots$$

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Linear filter model

- The basic model that we will use is of the neuron as a linear filter on the stimulus

$$r_{est}(t) = r_0 + \int_0^{\infty} D(\tau) s(t - \tau) d\tau$$

r_0 – baseline firing rate D – response kernel
 $r_{est}(t)$ – firing rate function $s(t)$ – stimulus function

- Rate → Convolution of the stimulus & response kernel.
- **What is the best rate estimation $r_{est}(t)$? Or in other words what is the best kernel $D_{opt}(t)$?**
- To do that we'll first define white noise...

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White noise stimulus

- White noise is random (non-restricted values), uncorrelated stochastic process - $s(t)$:

$$\langle s(t) \cdot s(t + \tau) \rangle = \sigma_s^2 \cdot \delta(\tau)$$
- The stimulus autocorrelation function - Q :

$$Q_{s,s}(\tau) = \frac{1}{T} \cdot \int_0^T s(t) \cdot s(t + \tau) dt$$
- Thus, the autocorrelation of white noise - Q :

$$Q_{s,s}(\tau) = \sigma_s^2 \cdot \delta(\tau)$$
- Assuming an ergodic signal we traded ensemble average for time average.

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Optimal kernel I

- The best rate estimator is defined by the difference from the actual rate function.

$$E = \frac{1}{T} \cdot \int_0^T (r_{est}(t) - r(t))^2 dt$$
- The best estimator minimizes the difference

$$\begin{aligned} \frac{dE}{dD(\tau)} &= \frac{2}{T} \int_0^T dt (r_{est}(t) - r(t)) s(t - \tau) \\ &= \frac{2}{T} \int_0^T dt \left(r_0 + \int_0^\infty d\tau' D(\tau') s(t - \tau') - r(t) \right) s(t - \tau) \\ &= 2 \int_0^\infty d\tau' D(\tau') Q_{ss}(\tau - \tau') d\tau' - Q_{rs}(-\tau) \end{aligned}$$

$$Q_{s,s}(\tau) = \frac{1}{T} \cdot \int_0^T s(t) \cdot s(t + \tau) dt \quad Q_{r,s}(\tau) = \frac{1}{T} \cdot \int_0^T r(t) \cdot s(t + \tau) dt$$

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Optimal kernel II

- The best estimation is achieved:

$$\int_0^\infty d\tau' Q_{ss}(\tau - \tau') D(\tau') = Q_{rs}(-\tau)$$

For white-noise stimuli $Q_{ss}(\tau) = \sigma_s^2 \delta(\tau)$, so

$$\sigma_s^2 \int_0^\infty \delta(\tau - \tau') D(\tau') d\tau' = \sigma_s^2 \cdot D(\tau) = Q_{r,s}(-\tau)$$

$$D(\tau) = \frac{Q_{rs}(-\tau)}{\sigma_s^2} = \frac{\langle r \rangle C(\tau)}{\sigma_s^2}$$

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Optimal kernel calculation

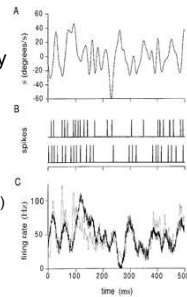
- Create white noise stimulus (no correlations).
- Calculate spike triggered average response to the stimulating white noise.
- Normalize the spike triggered average by the rate to get the optimal kernel.

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H1 neuron in visual system of blowfly

- Stimulus – image velocity
- Response of H1 neuron
- Estimated rate $r_{\text{est}}(t)$ (solid) assuming linear kernel
- Neural rate $r(t)$ (dashed) averaged spike trains



Panel A: Stimulus (image velocity) vs time (ms).
Panel B: Response of H1 neuron (spikes) vs time (ms).
Panel C: Firing rate (Hz) vs time (ms), showing estimated rate (solid) and neural rate (dashed).

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Optimal kernel without using white noise input

- Solving the equation: $\int_0^\infty d\tau Q_{ss}(\tau - \tau') D(\tau') = Q_{rs}(-\tau)$
- When the input is not white, a Fourier transform of the correlation functions enables finding the optimal kernel

$$D(\omega) = \frac{Q_{r,s}(-\omega)}{Q_{s,s}(\omega)}$$

$$D(\tau) = \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} \frac{Q_{r,s}(-\omega)}{Q_{s,s}(\omega)} \cdot e^{-i\omega\tau} d\omega$$

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Outline

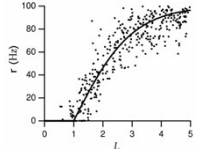
- Reverse correlations
- Linear filters
- Static non-linearity
- GLMs
- Appendix: V1 simple neurons

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Deviation from linearity

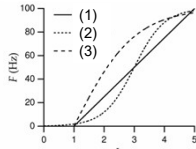
- Typically neurons deviate from the linear relationship: $r_{est}(t) = r_0 + \int D(\tau)s(t-\tau)d\tau = r_0 + L(t)$
- The most common deviations are **saturation** and **threshold**.



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Static non-linearity I



(1) $F(L) = G \cdot [L - 1]_+$

(2) $F(L) = \frac{r_{max}}{1 + e^{-2(L - L_0)}}$

(3) $F(L) = r_{max} \cdot [\tanh(0.5(L - L_0))]_+$

$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

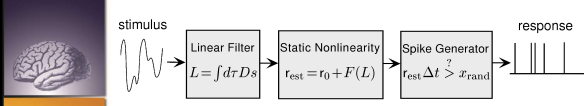
$G = 25 \text{ Hz}, L_0 = 1, L_{1/2} = 3, r_{max} = 100 \text{ Hz}, g_1 = 2, \text{ and } g_2 = 1/2$

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Static non-linearity II



The rate estimate is

$$r_{\text{est}}(t) = r_0 + F(L(t))$$

with F the non-linear function and L estimated using the linear theory.

A two-step approach: first estimate the optimal linear filter, then fit the best non-linearity.

Suboptimal: No inclusion of (non-linear interactions of) higher order moments
Linear filter is optimized ignoring the non-linearity.

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Non-linearity – other options

- The optimization method is non optimal even for static non linearity. However, *Bussgang* theorem demonstrates that it is close to optimal for Gaussian white noise.
- The other options are:
 - Optimize a non-linear function
- Use more terms in the Volterra or Wiener expansion

$$r_{\text{est}}(t) = r_0 + \int_0^\infty D(\tau) f(s(t-\tau)) d\tau$$

$$r_{\text{est}}(t) = r_0 + \int_0^\infty d\tau D(\tau) s(t-\tau) + \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 D_2(\tau_1, \tau_2) s(t-\tau_1) s(t-\tau_2) + \dots$$

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Outline

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GLMs

- General linear model
- Generalized linear model

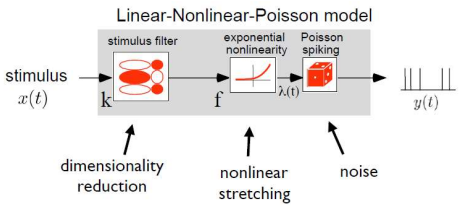
(Slides of this section are adapted from Jonathan Pillow)

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“Basic” Poisson GLM

Linear-Nonlinear-Poisson model



stimulus $x(t)$ → k → f → $\lambda(t)$ → Poisson spiking → $y(t)$

dimensionality reduction nonlinear stretching noise

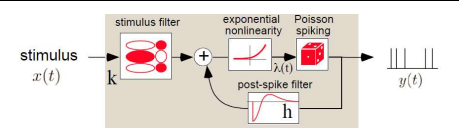
spike rate $\lambda = f(\vec{k} \cdot \vec{x})$

spike count $y \sim \text{Pois}(\lambda)$

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History dependent GLM



stimulus $x(t)$ → k → $+$ → $\lambda(t)$ → Poisson spiking → $y(t)$

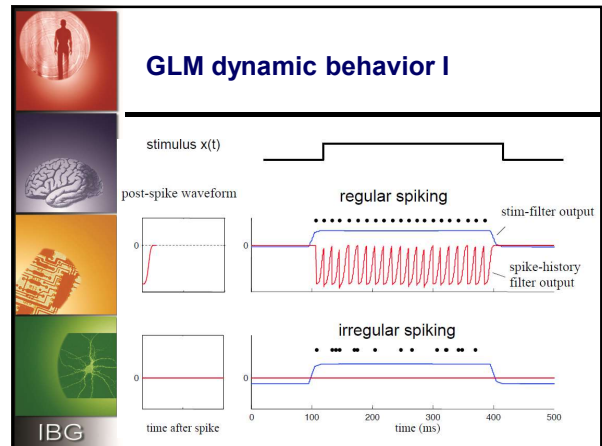
post-spike filter h → $+$

conditional intensity (spike rate) $\lambda(t) = f(k \cdot x(t) + h \cdot y(t))$

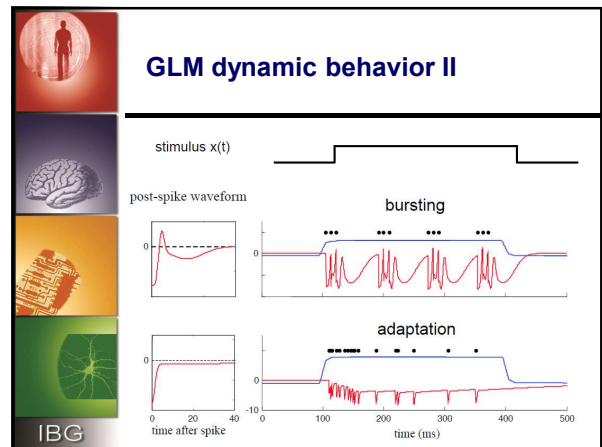
$= e^{k \cdot x(t)} \cdot e^{h \cdot y(t)}$

- rate is a product of stim- and spike-history dependent terms
- output no longer a Poisson process
- also known as “soft-threshold” Integrate-and-Fire model

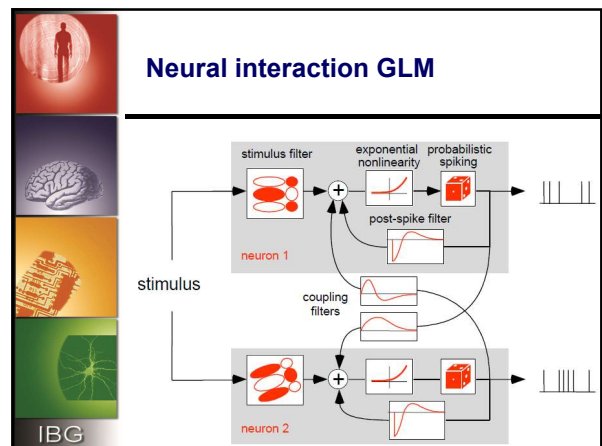
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Outline

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Visual stimuli

Sinusoidal grating

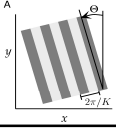
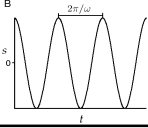
$$s(x, y, t) = A \cos(Kx \cos \Theta + Ky \sin \Theta - \Phi) \cos(\omega t)$$

K, ω spatial, temporal frequency. Θ is orientation.

White-noise stimulus

$$\langle s(x, y, t) s(x', y', t') \rangle = \sigma^2 \delta(t - t') \delta(x - x') \delta(y - y')$$

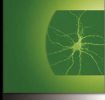
$\langle s \rangle = 0$ to avoid dependence on overall illumination.

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Spatial receptive fields

We generalize previous concepts to 2-d visual stimuli $s(x, y, t)$:

Spike-triggered average $C(x, y, \tau) = \frac{1}{n} \langle \sum_{i=1}^n s(x, y, t_i - \tau) \rangle$

Correlation function $Q_{rs}(x, y, \tau) = \frac{1}{T} \int_0^T dt r(t) s(x, y, t + \tau)$

$$C(x, y, \tau) = \frac{Q_{rs}(x, y, -\tau)}{\langle r \rangle}$$

Linear filter $L(t) = \int_0^\infty d\tau \int dx dy D(x, y, \tau) s(x, y, t - \tau)$

For white-noise $D(x, y, \tau) = \frac{Q_{rs}(x, y, -\tau)}{\sigma_s^2}$

Separable kernel $D(x, y, \tau) = D_s(x, y) D_t(\tau)$

$$D_s(x, y) \propto \int d\tau D(x, y, \tau)$$

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V1 spatial receptive fields



$D(x, y)$ from spike triggered average of two different cat visual cortex area 17 simple cells.



Stimulus is averaged 50-100 msec prior to action potential.



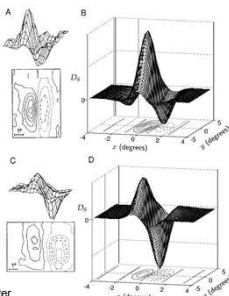
$D(x, y)$ shows separate ON and OFF region.

Simple cells with up to 5 regions are found.

Gabor function

$$D(x, y) = \frac{\cos(kx - \phi)}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right)$$

(Border parallel to y axis i.e. $\theta=0$, origin at center)



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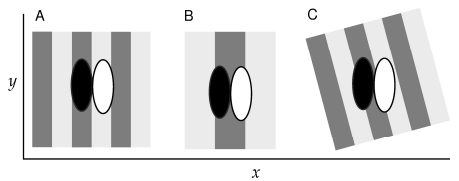


Response to grating









A: Grating stimuli superimposed on spatial receptive fields.
 • Dark oval – OFF area $D_s < 0$, White oval – ON area $D_s > 0$
 • Optimal response when both spatial frequency and orientation of stimulus and filter match (such as in figure A)

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Gabor functions





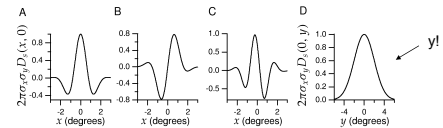


Examples of other shapes of Gabor functions:

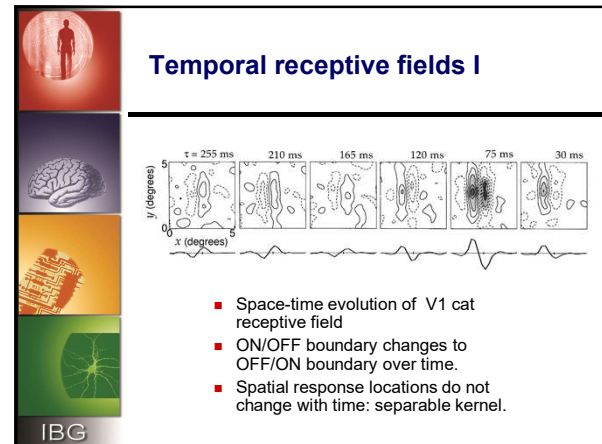
$$D_s(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \exp\left(-\frac{x^2}{2\sigma_x^2} - \frac{y^2}{2\sigma_y^2}\right) \cos(kx - \phi)$$

Preferred orientation of light bars is parallel to the y direction.

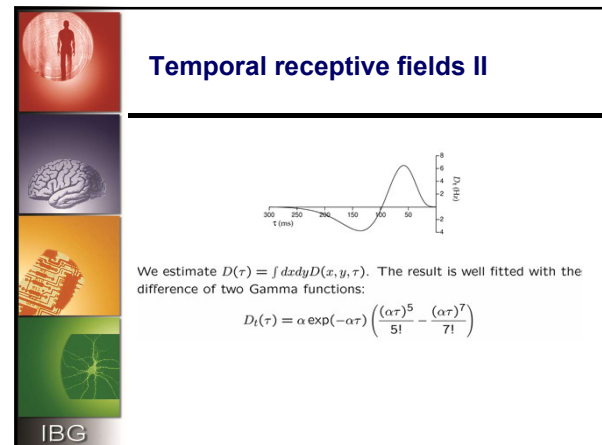
A: $D(x, 0)$ vs. x with $\sigma_x = 1^0, 1/k = 0.5^0, \phi = 0$
 B: $D(x, 0)$ vs. x with $\sigma_x = 1^0, 1/k = 0.5^0, \phi = \pi/2$
 C: $D(x, 0)$ vs. x with $\sigma_x = 1^0, 1/k = 0.33^0, \phi = \pi/4$
 D: $D(0, y)$ vs. y with $\sigma_y = 2^0$



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